

Engineering Curves: Construction of Epicycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Given

A generating circle of diameter 30 mm rolls externally without slipping on a directing circle of diameter 90 mm.

- Construct one complete epicycloid arch. $\angle AOB = 120^\circ$ and the arch is divided into 12 equal angular parts. Retain every construction element in all later views.

Calculated Data

$$R = 90/2 = 45 \text{ mm},$$

$$OC_0 = R + r = 60 \text{ mm},$$

$$\angle AOB = 120^\circ,$$

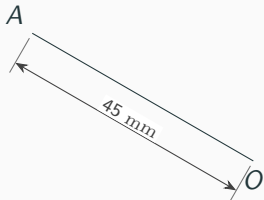
$$r = 30/2 = 15 \text{ mm},$$

$$AC_0 = 15 \text{ mm},$$

$$\Delta\theta = 120^\circ/12 = 10^\circ.$$

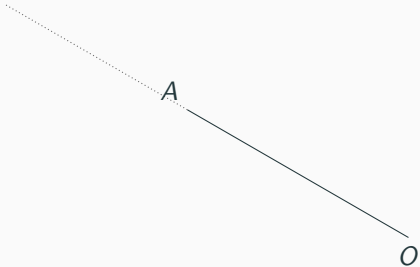
Step 1: Draw OA

$OA = 45 \text{ mm.}$



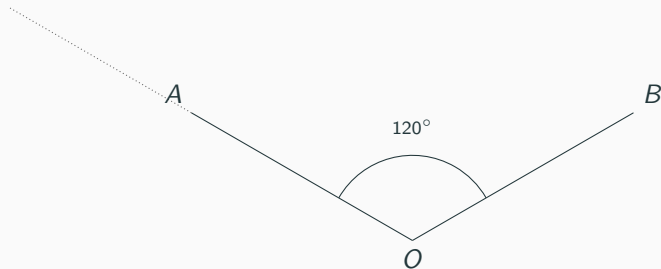
Step 2: Extend OA

Extend OA .



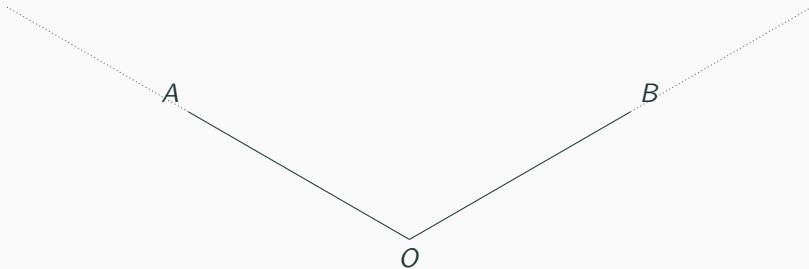
Step 3: Draw OB at 120°

$$\angle AOB = 120^\circ.$$



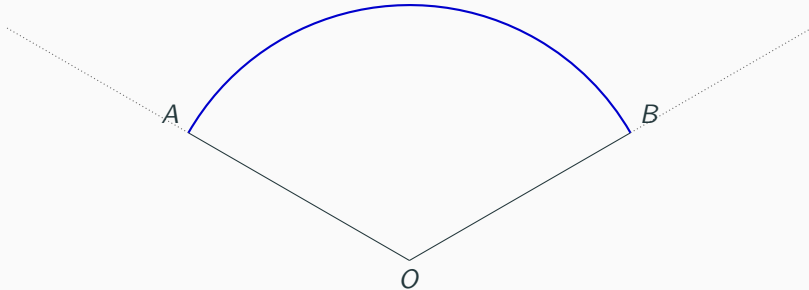
Step 4: Extend OB

Extend OB .



Step 5: Draw the directing arc AB

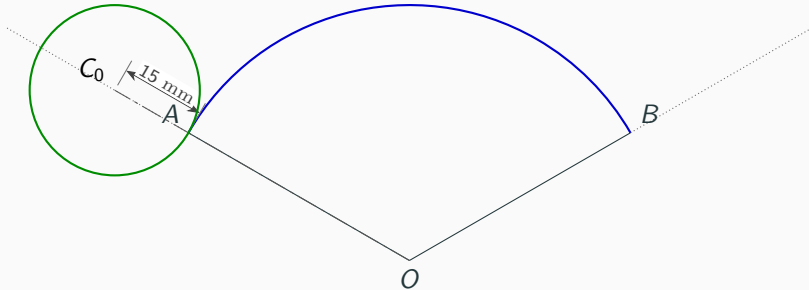
Blue arc AB : directing circle.



Step 6: Locate C_0 and the generating circle

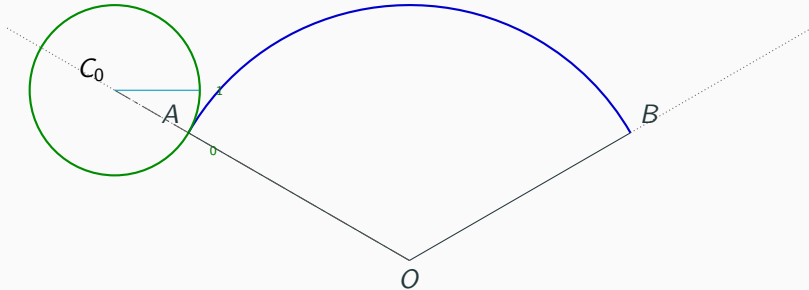
$AC_0 = 15 \text{ mm.}$

Green: generating circle.



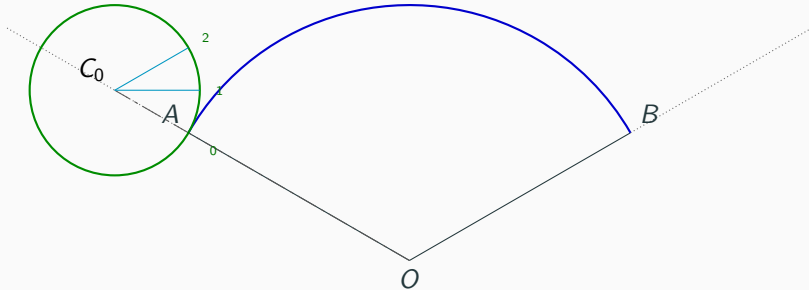
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



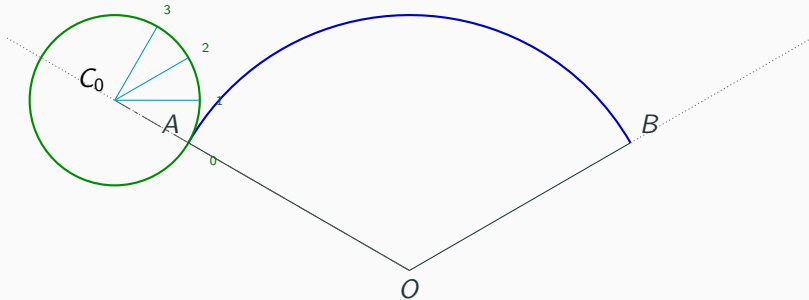
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



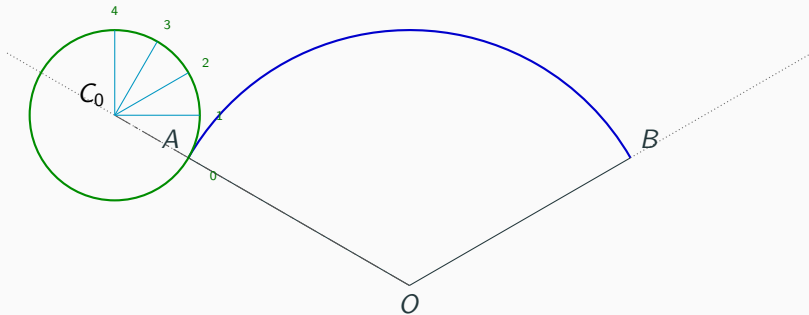
Step 7: Divide the generating circle into 12 equal parts

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30° divisions.



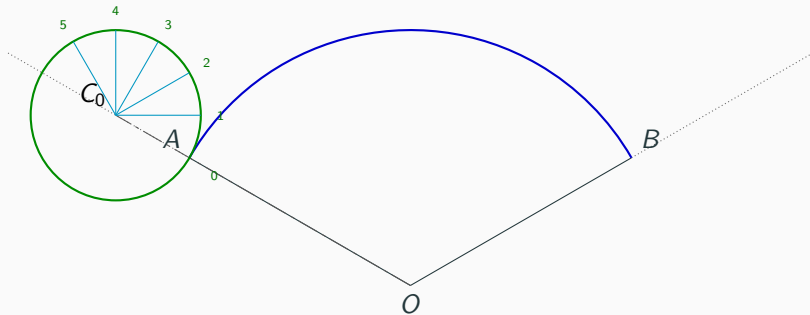
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



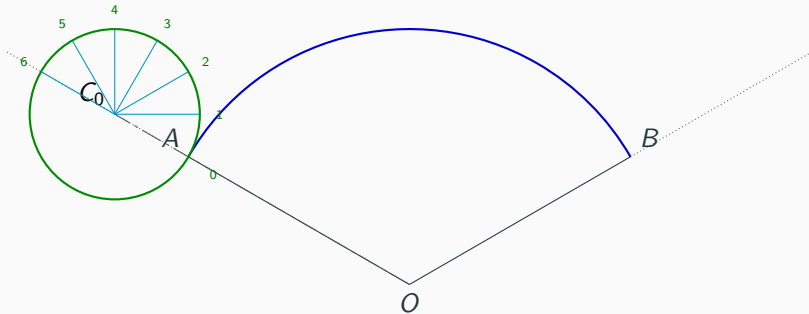
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



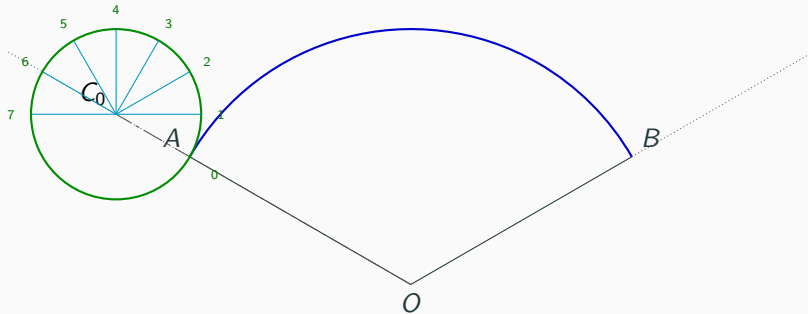
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



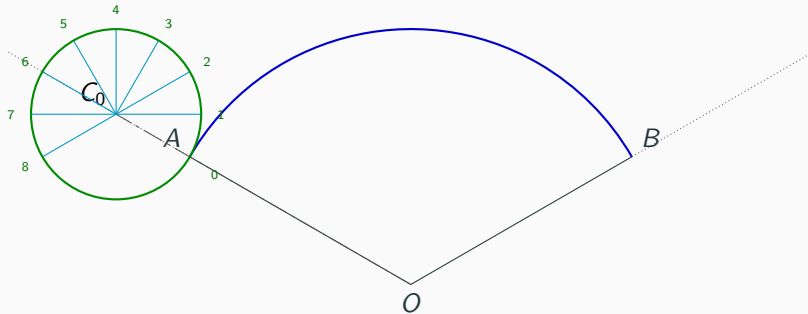
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



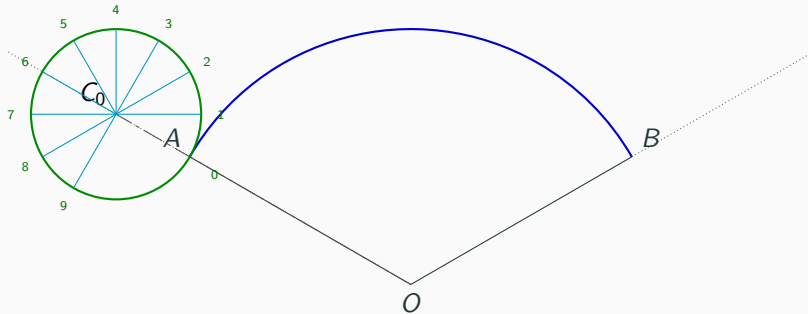
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



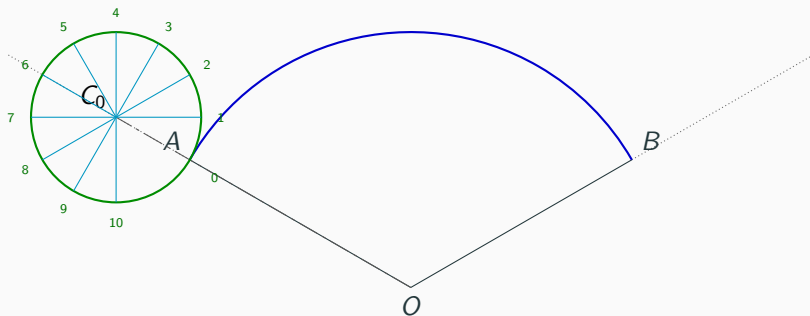
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



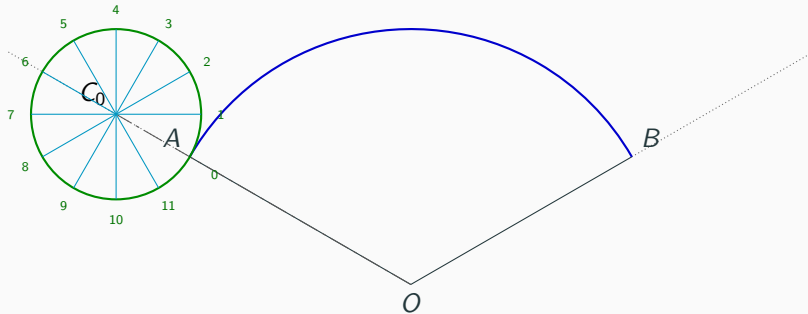
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



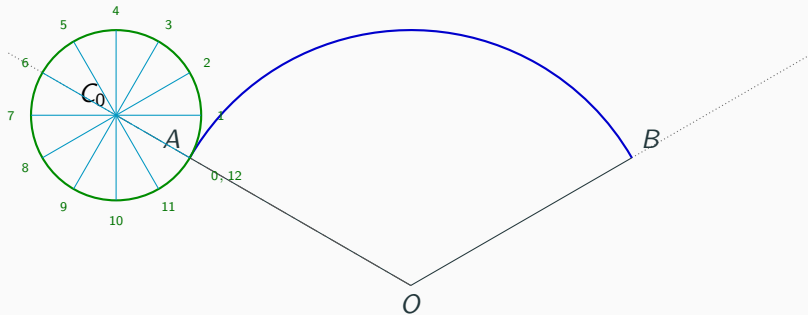
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



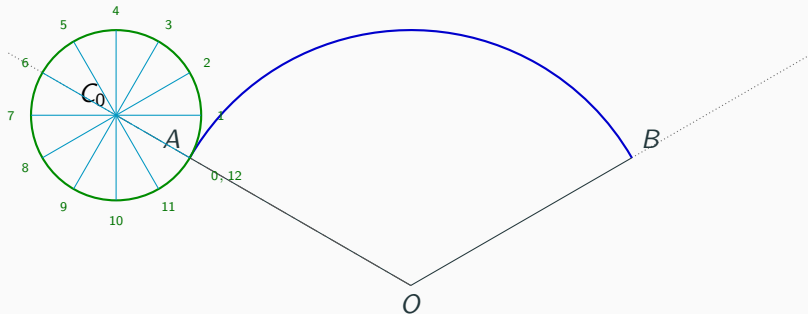
Step 7: Divide the generating circle into 12 equal parts

Green: generating circle.
30° divisions.



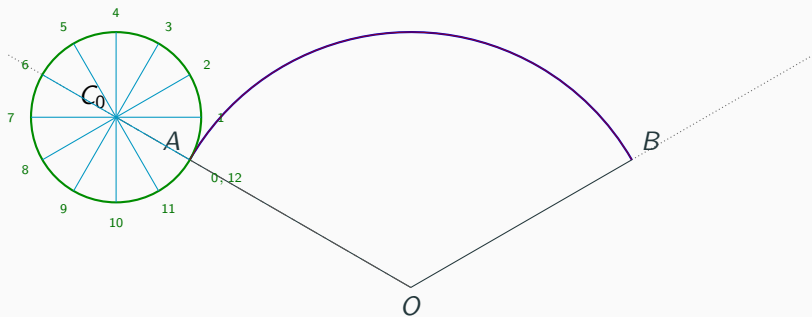
Step 8: Complete the generating-circle divisions

Generating-circle divisions.



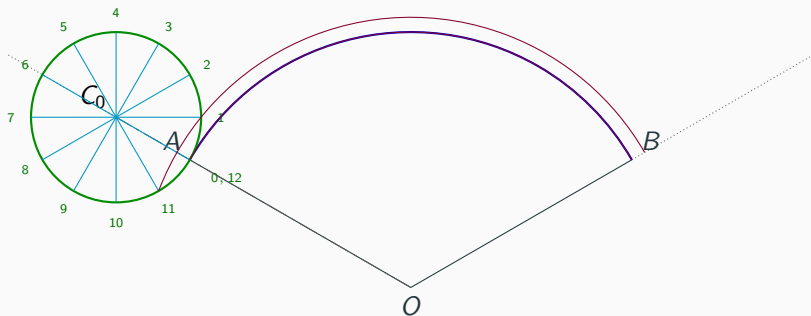
Step 9: Draw concentric construction arcs

Concentric construction arc.



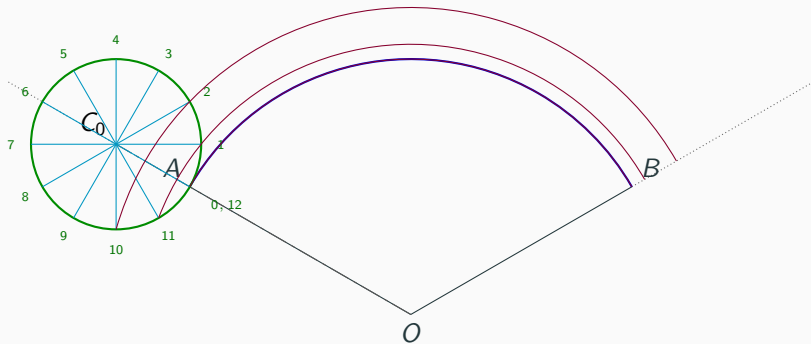
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Concentric construction arc.



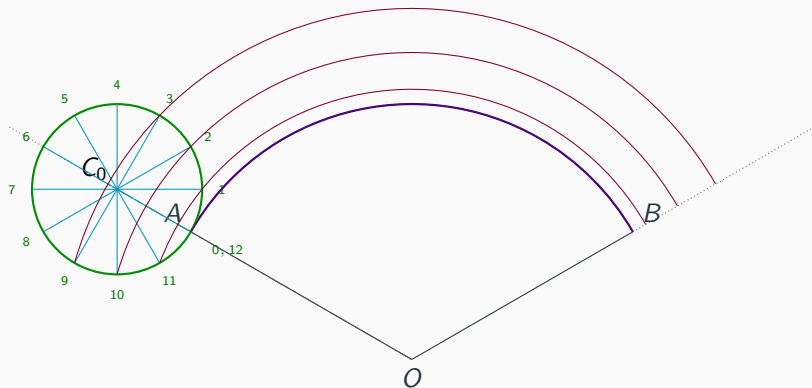
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Concentric construction arc.



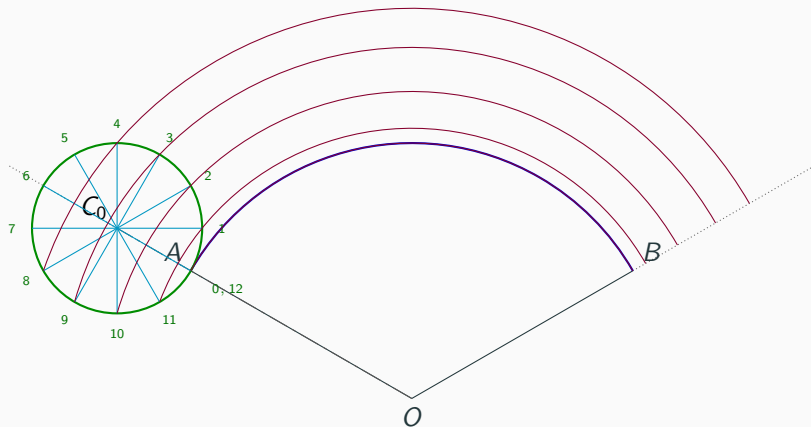
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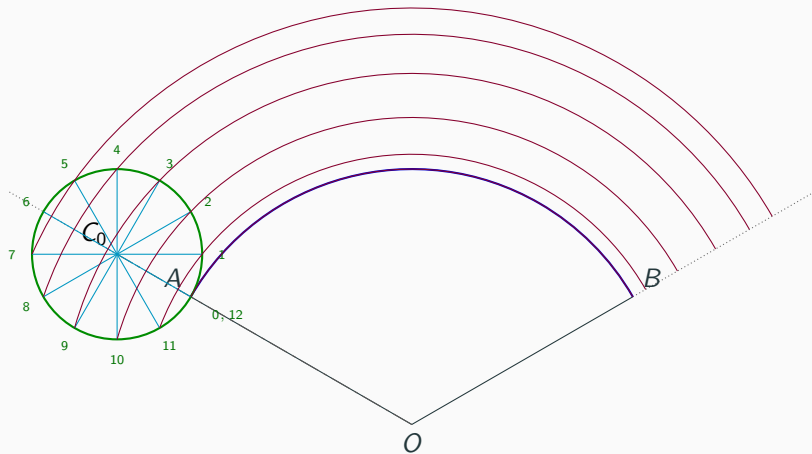
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Concentric construction arc.



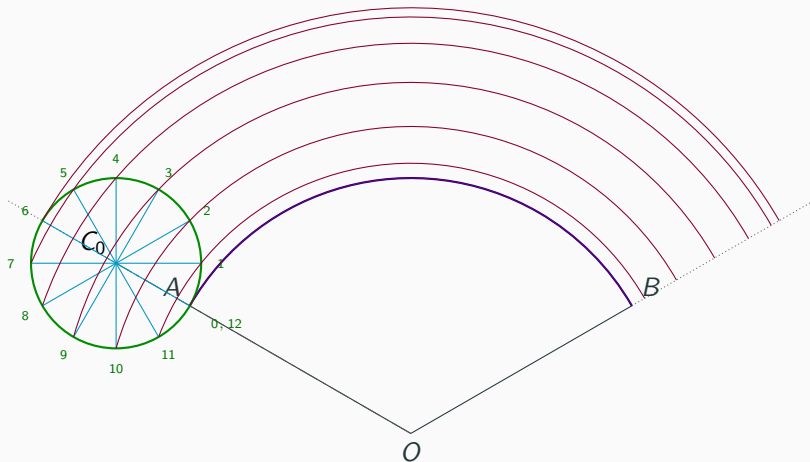
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Concentric construction arc.

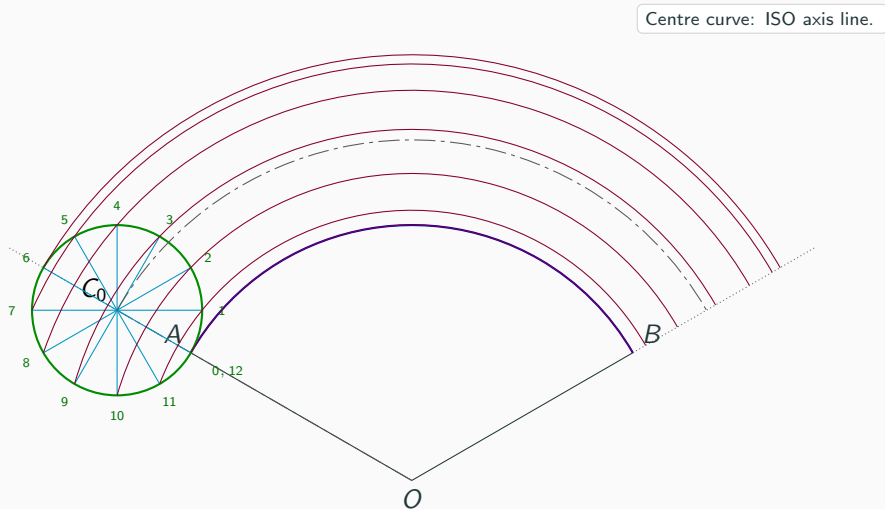


Step 9: Draw concentric construction arcs

Concentric construction arc.

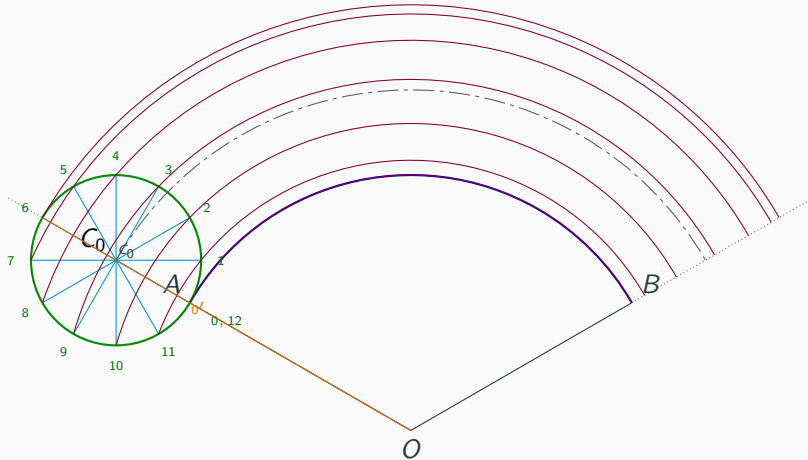


Step 10: Draw the centre arc through C_0



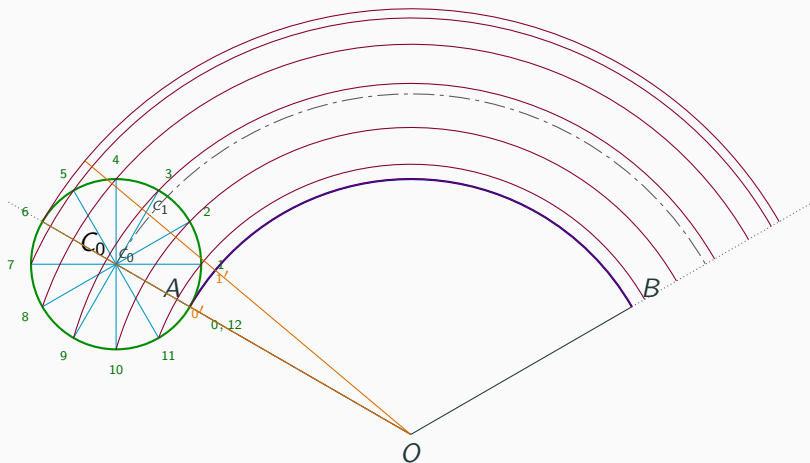
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



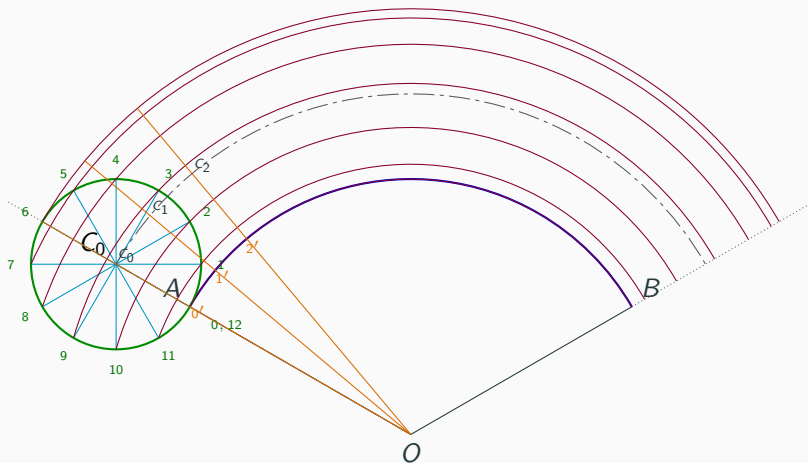
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



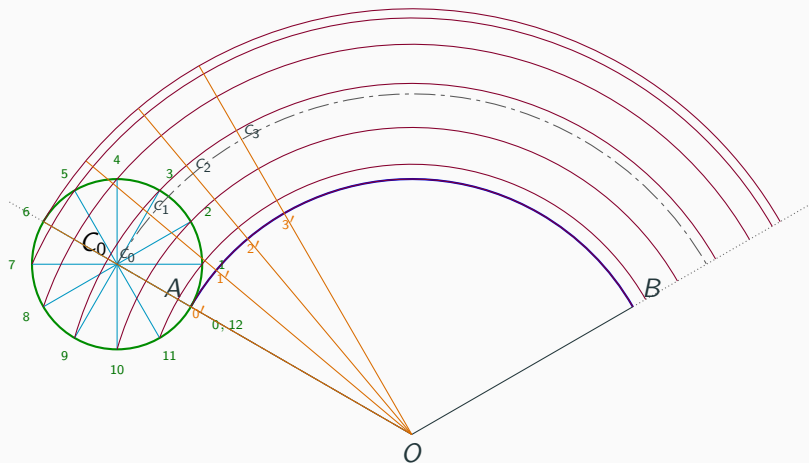
Step 11: Divide $\angle AOB$ into 12 equal parts

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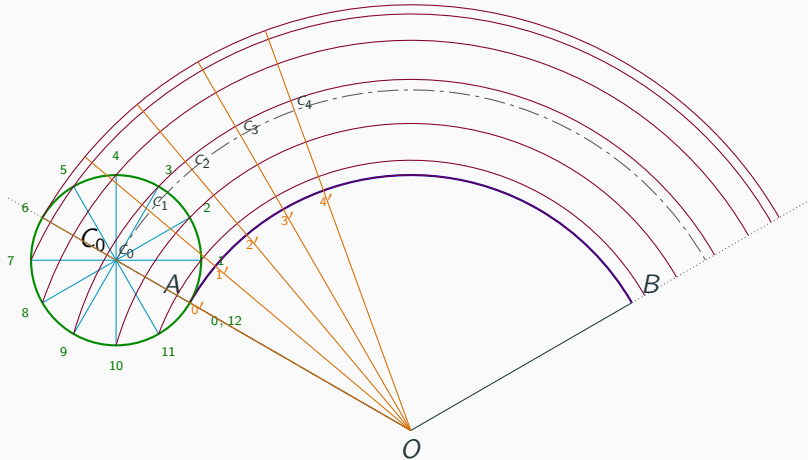
Step 11: Divide $\angle AOB$ into 12 equal parts

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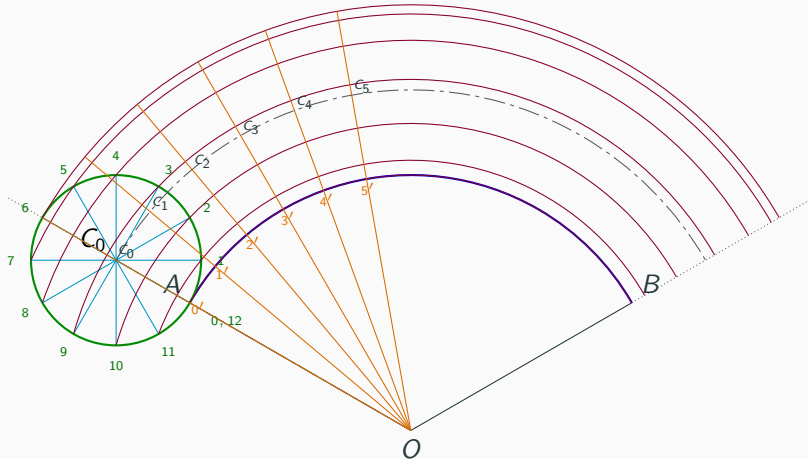
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



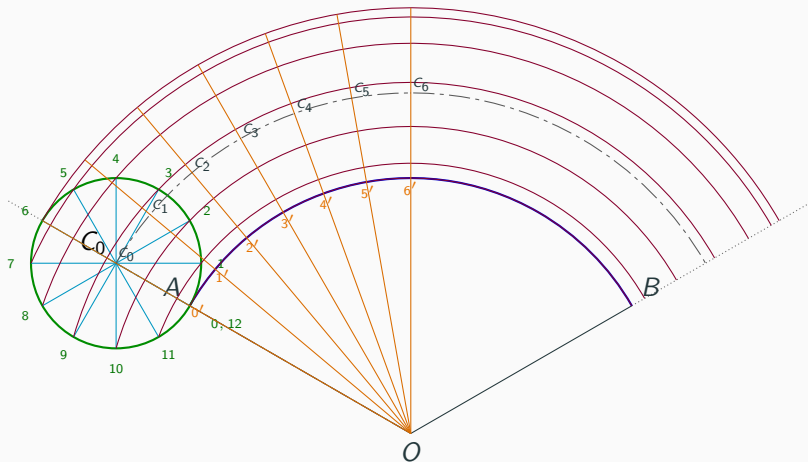
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



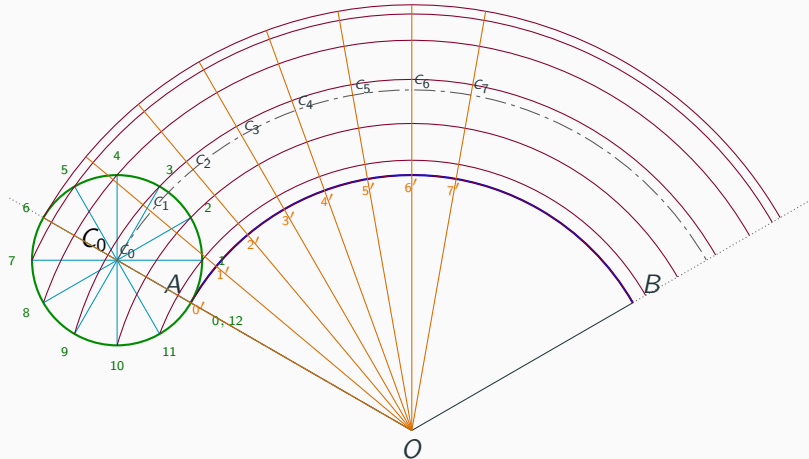
Step 11: Divide $\angle AOB$ into 12 equal parts

10^0 divisions.



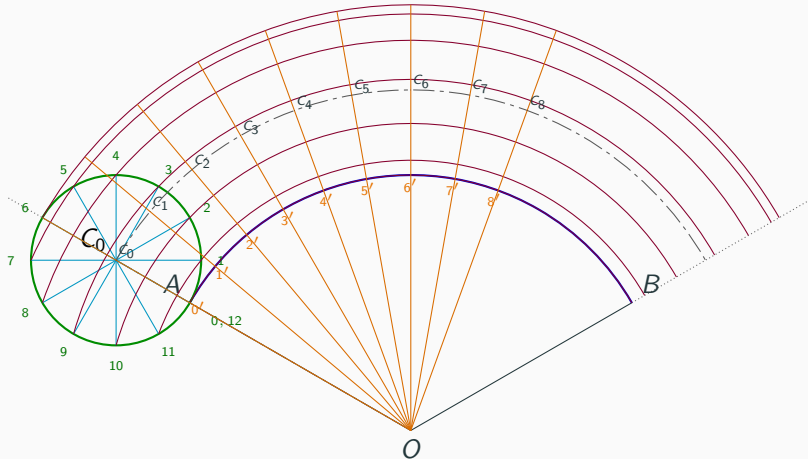
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



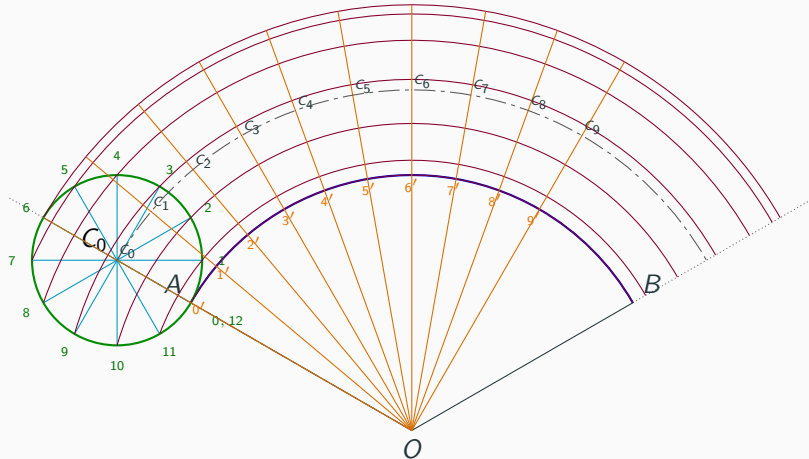
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



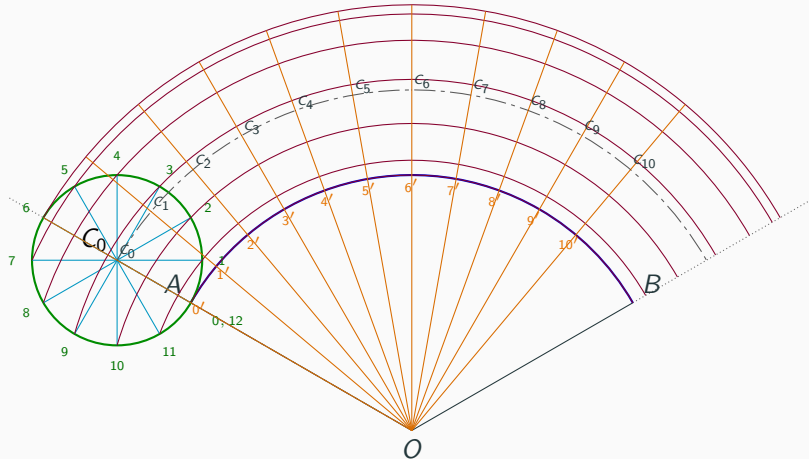
Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.



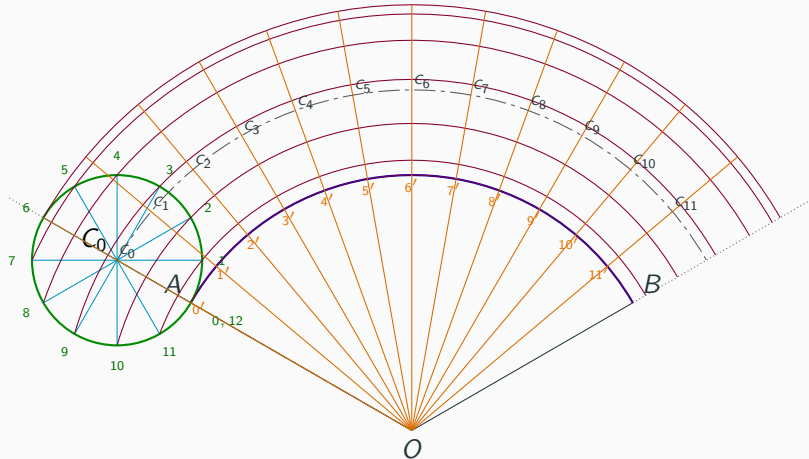
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10° divisions.



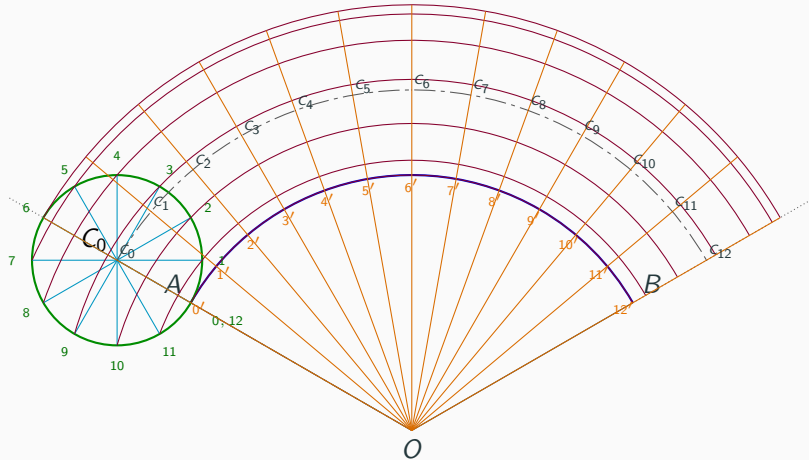
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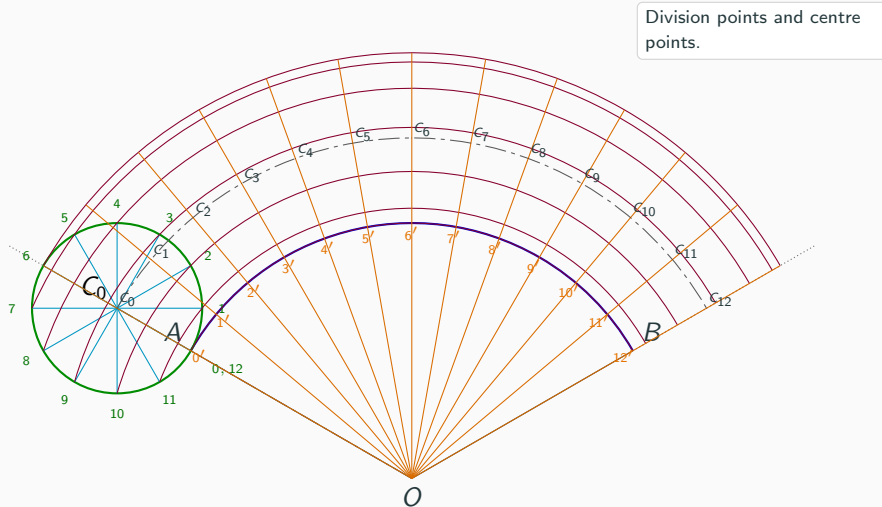


Step 11: Divide $\angle AOB$ into 12 equal parts

10° divisions.

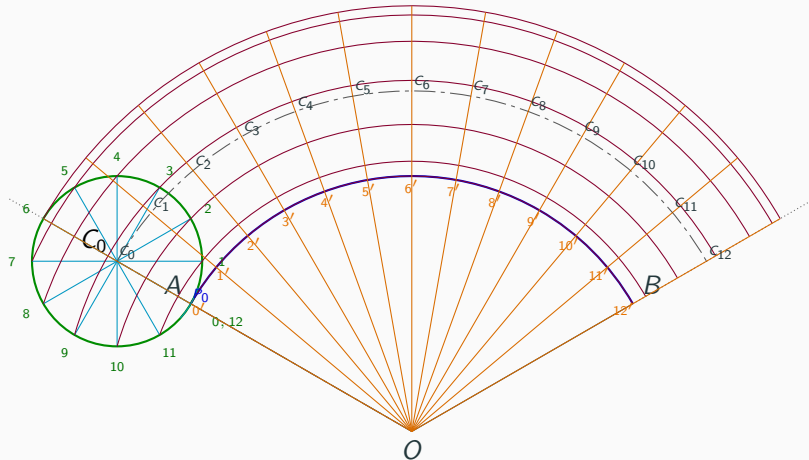


Step 12: Complete the angular divisions and points



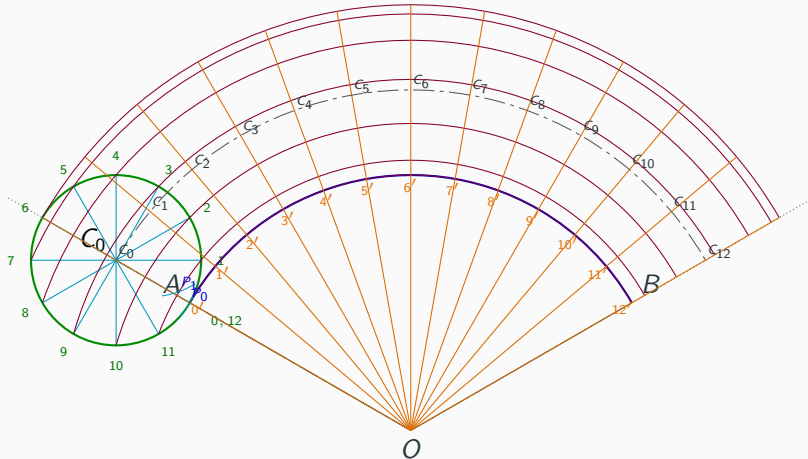
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



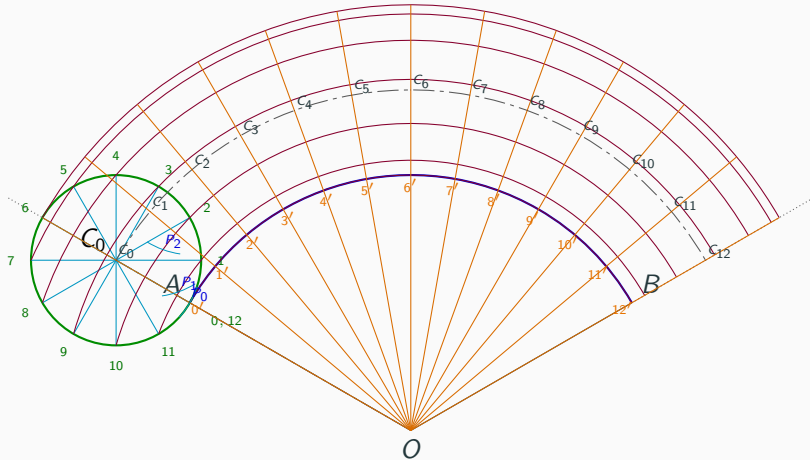
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



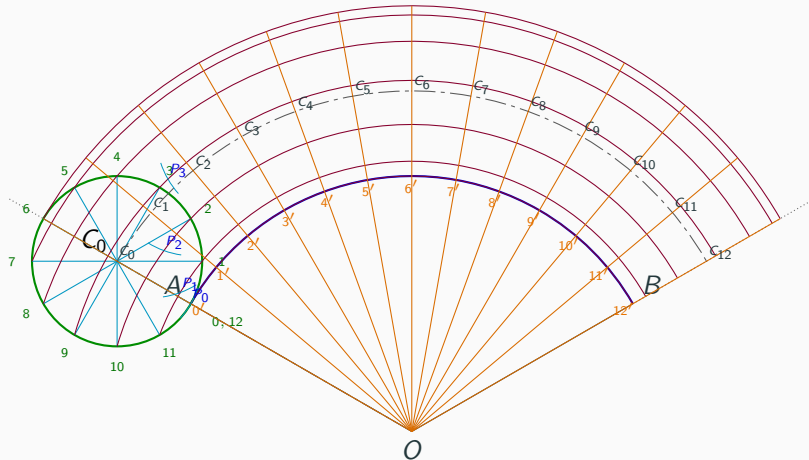
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



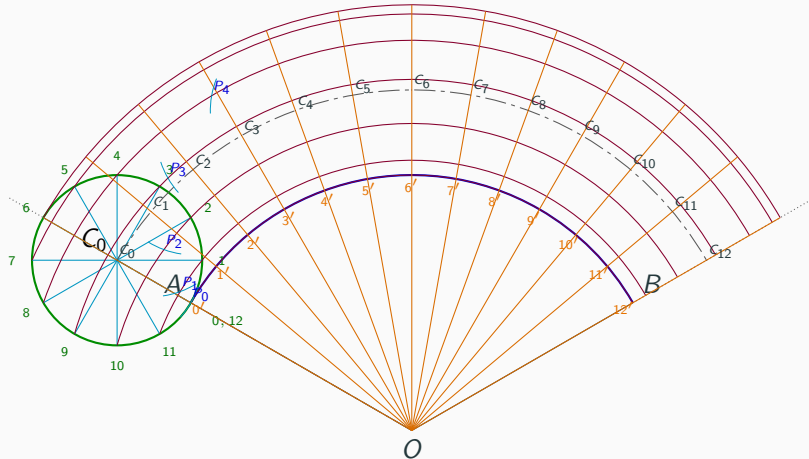
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.



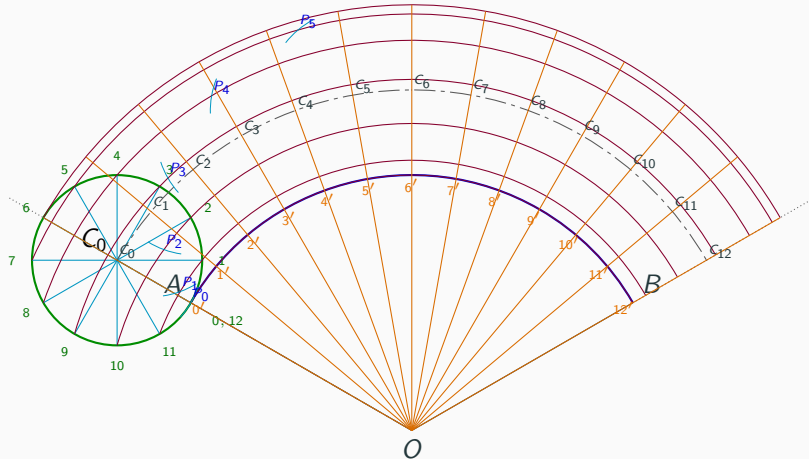
Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

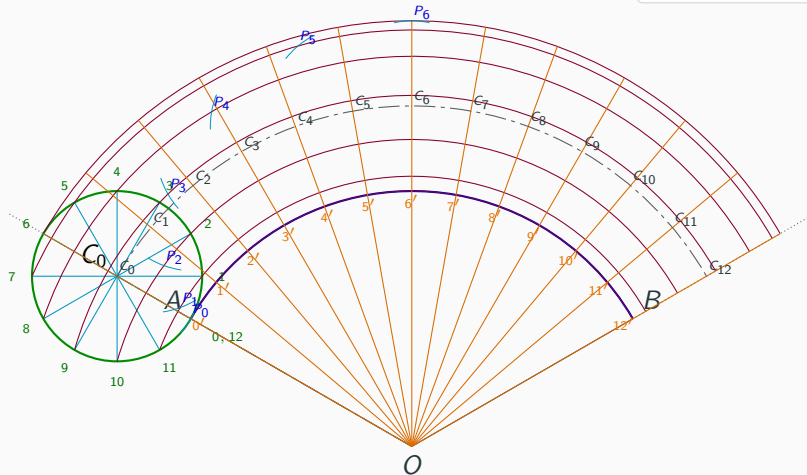


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

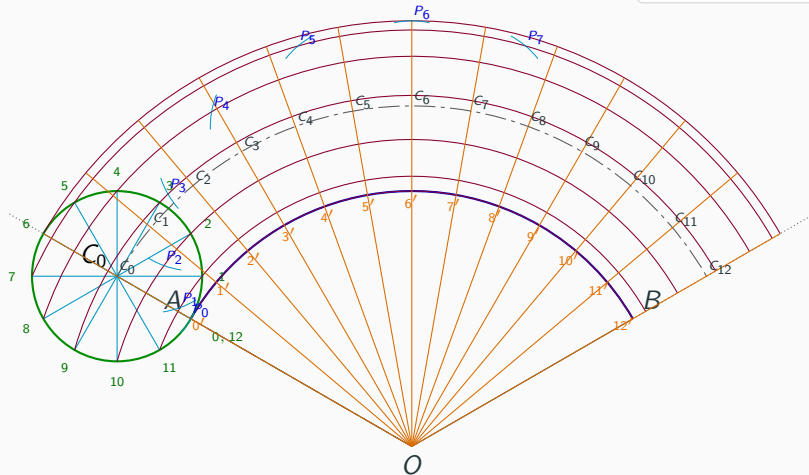


Step 13: Draw the small cutting arc and locate P_i



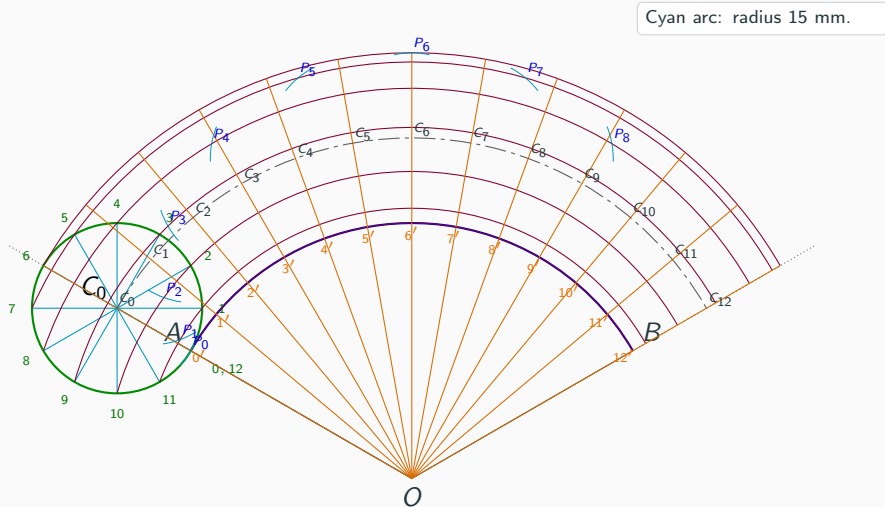
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i



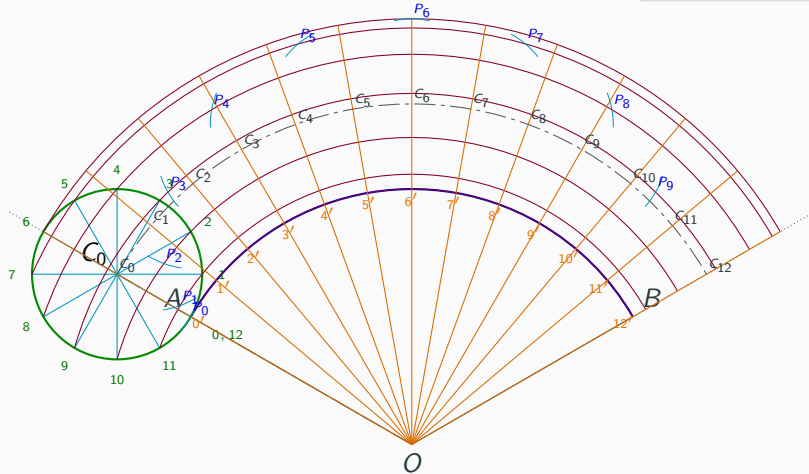
Cyan arc: radius 15 mm.

Step 13: Draw the small cutting arc and locate P_i

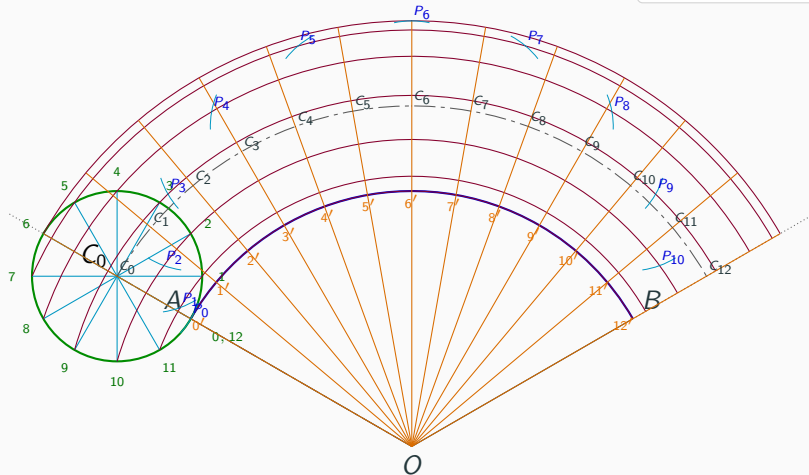


Step 13: Draw the small cutting arc and locate P_i

Cyan arc: radius 15 mm.

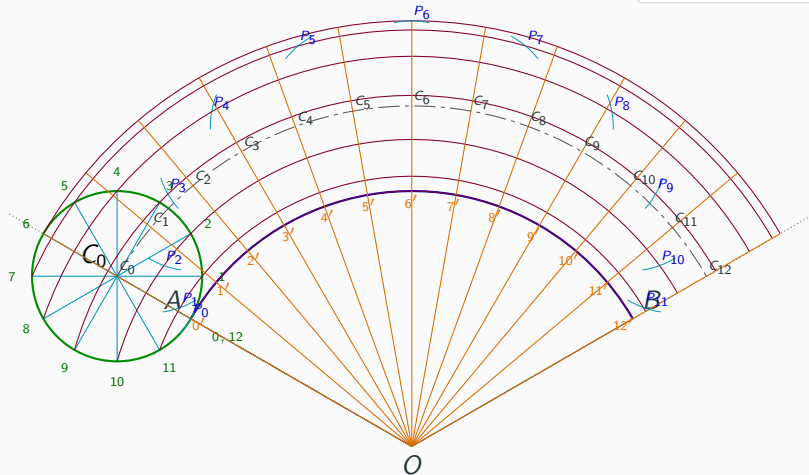


Step 13: Draw the small cutting arc and locate P_i

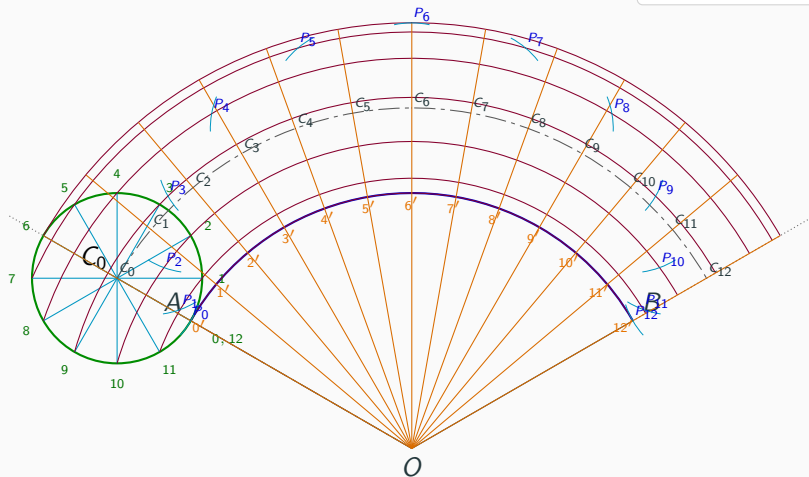


Cyan arc: radius 15 mm.

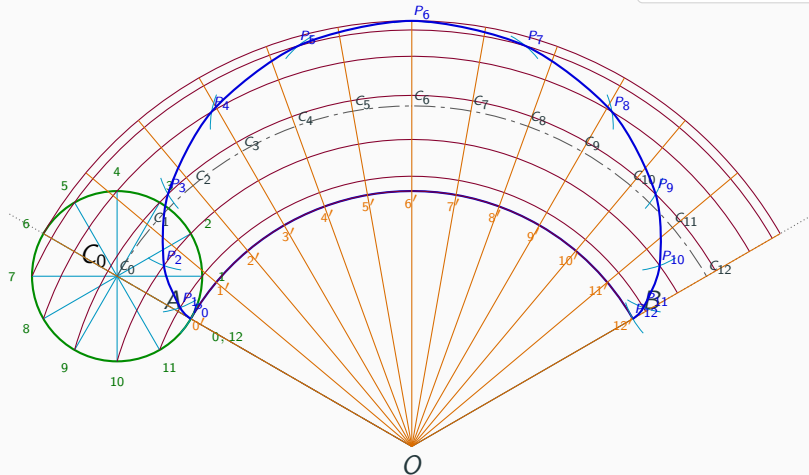
Step 13: Draw the small cutting arc and locate P_i



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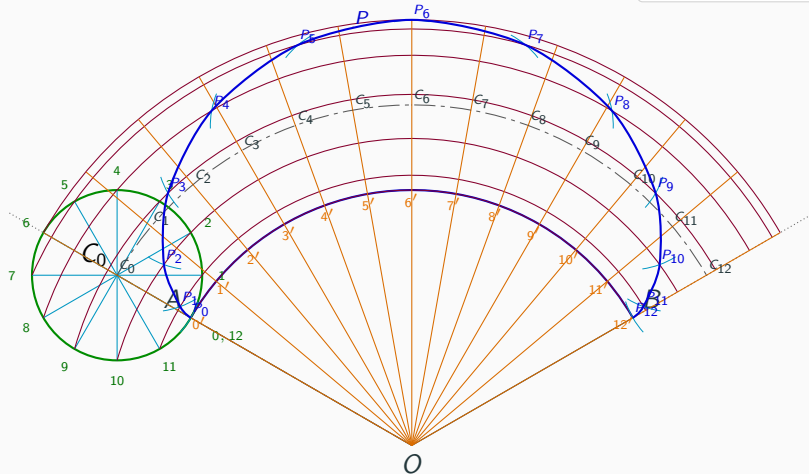


Step 14: Draw the epicycloid

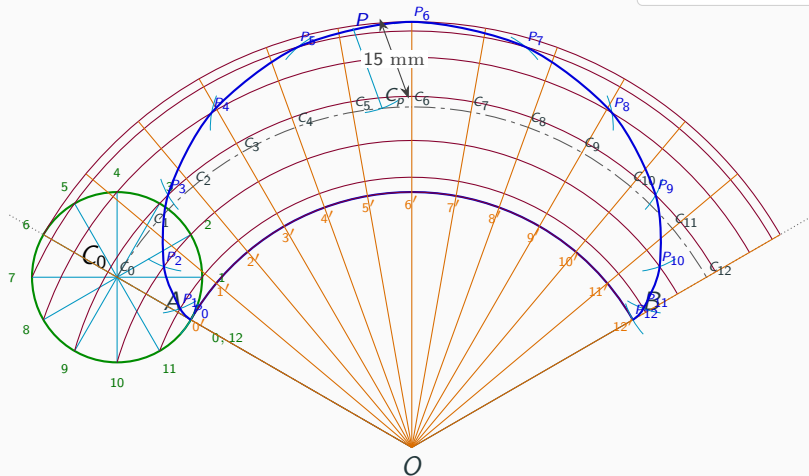


Blue curve: epicycloid.

Step 15: Select a point P on the epicycloid

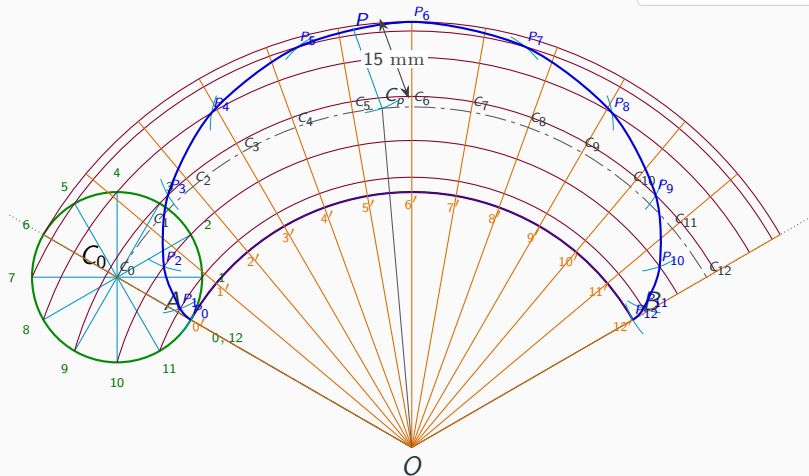


Step 16: Draw the 15 mm arc from P to the centre curve



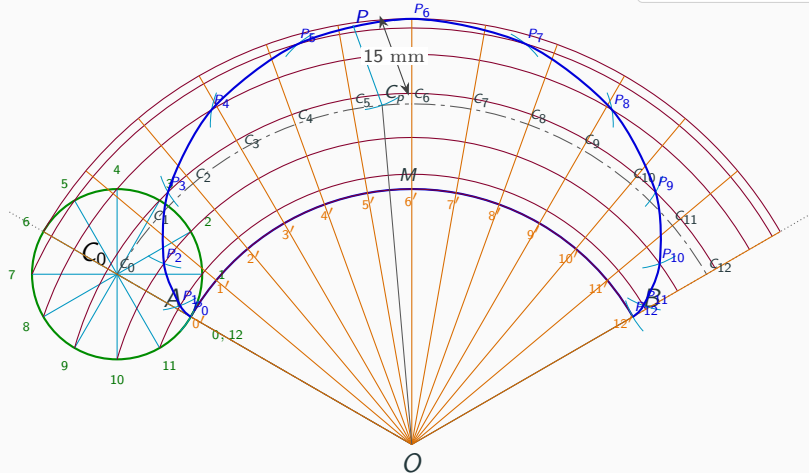
$$PC_p = 15 \text{ mm}$$

Step 17: Join C_p to O

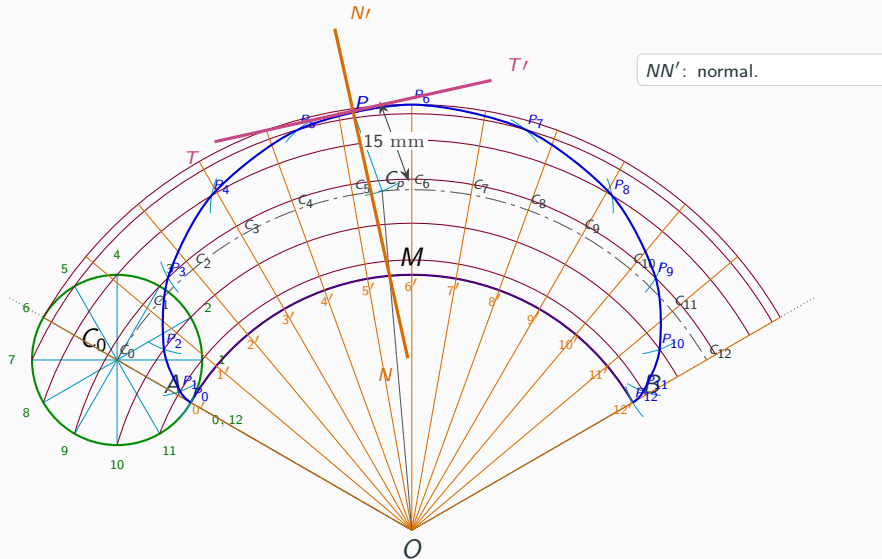


Step 18: Mark M on the directing circle

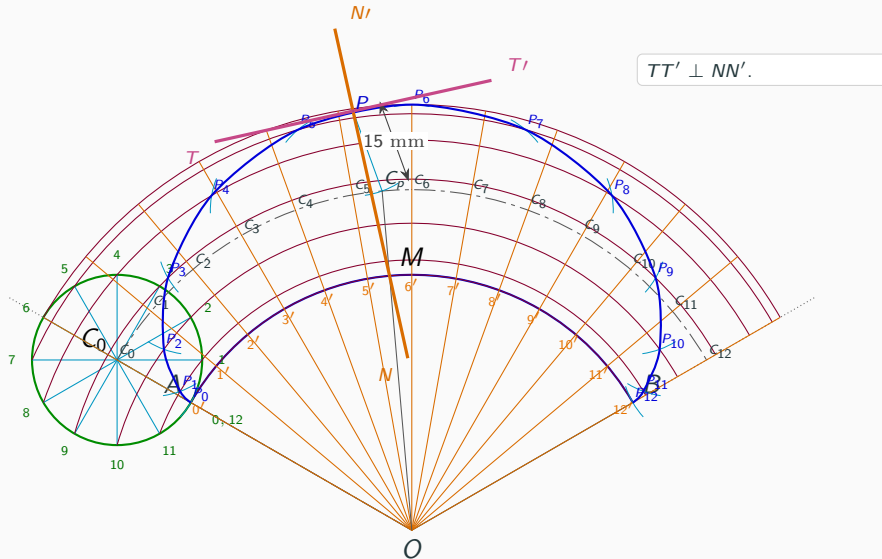
$$M = OC_p \cap AB.$$



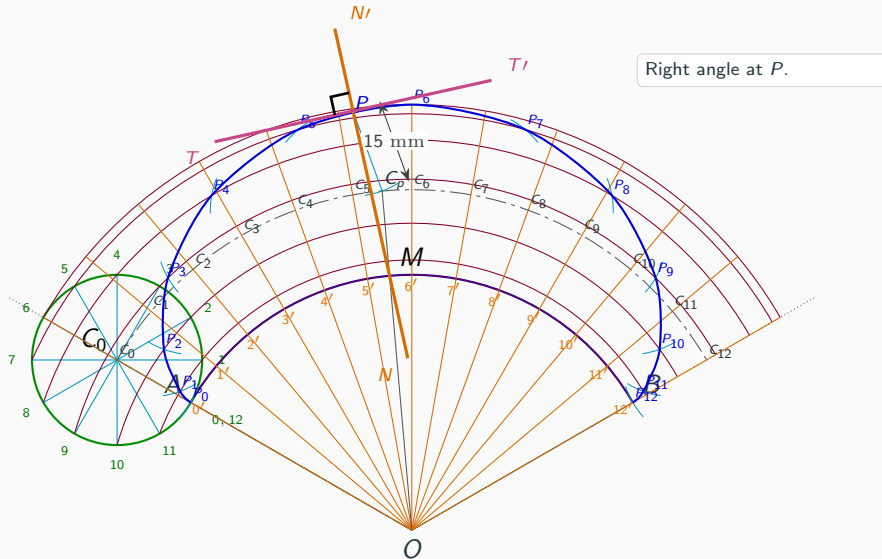
Step 19: Draw the normal NN' through P



Step 20: Draw the tangent perpendicular to NN'



Step 20: Show the right-angle at P



Step 20: Final completed epicycloid construction

